

PROBABILITY & PROBABILITY DISTRIBUTIONS

Complete Textbook Guide: Deep Core Concepts & Structured Solutions
Matrix

Exclusions: Zero Previous Year Questions (No Historical PYQ Content)

Inclusions: Axiomatic Mechanics, Random Variables, Discrete & Continuous Probability Distributions

Target Standard: Academic Core & Advanced Technical Frameworks — Proper English Copy

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CHAPTER 1: FOUNDATIONS OF PROBABILITY & RANDOM VARIABLES

1.1 Random Variables: Mapping Realities to Scalar Space

A random variable X is not a static algebraic unknown, but rather a mathematically precise real-valued function that maps every unique element of an experimental sample space $\mathcal{S}$ onto a continuous or discrete real number coordinate system. Random variables are categorized into two fundamentally distinct types depending on the nature of their range set:

- **Discrete Random Variables:** Can assume only isolated values along a countable list (e.g., integers). Their mapping behavior is governed by a **Probability Mass Function (PMF)**, denoted as $p(x) = P(X = x)$, subject to the conditions that $p(x) \geq 0$ and $\sum p(x) = 1$.
- **Continuous Random Variables:** Can take any real numerical value across a smooth, continuous interval scale. Since hitting an exact coordinate has a probability of zero, these variables are governed by a **Probability Density Function (PDF)**, denoted as *The Cumulative Distribution Function (CDF)*, denoted universally as $F(x)$, defines the total accumulated probability mass up to a terminal upper boundary coordinate x :

$$F(x) = P(X \leq x)$$

For a continuous variable, $F(x) = \int_{-\infty}^x f(t)dt$. The function is strictly monotonically non-decreasing, bounded cleanly such that $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 1$.

The Binomial distribution models a system consisting of a fixed number n of independent and identical Bernoulli trials. Each trial must produce exactly two mutually exclusive outcomes: success (with

CHAPTER 2: THEORY OF DISCRETE PROBABILITY DISTRIBUTIONS

2.1 The Binomial Distribution (Bernoulli Array)

constant probability p) or failure (with probability $q = 1 - p$). The probability mass function of getting exactly k successes is defined as:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k \in \{0, 1, 2, \dots, n\}$$

The structural properties of a Binomial space are completely defined by its first two parameters:

$$\text{ext{Mean}} (\mu) = np \quad | \quad \text{ext{Variance}} (\sigma^2) = np(1-p)$$

2.2 The Poisson Distribution (The Law of Rare Events)

The Poisson distribution arises as the limiting case of a Binomial distribution when the number of trials approaches infinity ($n \rightarrow \infty$) and the success probability approaches zero ($p \rightarrow 0$), while their product remains constant as a stable rate parameter ($\lambda = np$). It models the number of events occurring within a fixed interval of time or space:

$$P(X = k) = \frac{e^{-\lambda}}{k!} \cdot \lambda^k, \quad k \in \{0, 1, 2, \dots\}$$

A unique physical characteristic of Poisson space is that its dispersion equals its central balance point:

$$\text{ext{Mean}} (\mu) = \lambda \quad | \quad \text{ext{Variance}} (\sigma^2) = \lambda$$

CHAPTER 3: THEORY OF CONTINUOUS PROBABILITY DISTRIBUTIONS

3.1 The Uniform (Rectangular) Distribution

The continuous Uniform distribution represents a state of complete uncertainty over a bounded interval $[a, b]$. The probability density remains perfectly constant across the entire range, meaning the probability is directly proportional to the length of the sub-interval:

$$f(x) = \frac{1}{b - a} \quad \text{for } a \leq x \leq b, \quad \text{and } f(x) = 0 \text{ elsewhere}$$

Its central balance coordinate and spatial dispersion are derived as:

$$\text{Mean } (\mu) = \frac{a + b}{2} \quad \text{and} \quad \text{Variance } (\sigma^2) = \frac{(b - a)^2}{12}$$

3.2 The Normal / Gaussian Distribution (The Bell Curve Continuum)

By the Central Limit Theorem, when a large number of independent random variables are added together, their normalized sum tends toward a Gaussian distribution, regardless of the underlying shape of the individual distributions. This makes the Normal distribution the most vital continuous curve in statistics. It is defined by its mean μ and variance σ^2 .

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} \cdot e^{-\frac{1}{2} \left(\frac{x - \mu}{\sigma} \right)^2}, \quad -\infty < x < \infty$$

To evaluate areas under this curve without computing complex integrals every time, we transform the variable into standard coordinate space using the **Standard Normal Transformation Z-score**

$$Z = \frac{X - \mu}{\sigma} \implies f(z) = \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{z^2}{2}}$$

CHAPTER 4: CORE TEXTBOOK PROBLEM BANK (WITH DETAILED SOLUTIONS)

Section 4.1: Discrete Distribution Problems

Problem 1: Variance Derivation Matrix of a Poisson Distribution

Using the formal definition of mathematical expectation for discrete variables, prove from scratch that the variance of a Poisson random variable with rate parameter λ is identically equal to λ .

Detailed Step-by-Step Proof:

1. The algebraic identity for variance is: $\text{Var}(X) = E[X^2] - (E[X])^2$. We know for a Poisson distribution that $E[X] = \lambda$.

2. To find $E[X^2]$ smoothly without complicated summation series, rewrite the squared term as factorial components: $X^2 = X(X-1) + X$.

3. Apply the linearity property of expectations: $E[X^2] = E[X(X-1)] + E[X] = E[X(X-1)] + \lambda$.

4. Write out the explicit summation formula for $E[X(X-1)]$ using the Poisson PMF:

$$E[X(X-1)] = \sum_{k=0}^{\infty} k(k-1) \frac{e^{-\lambda} \lambda^k}{k!}$$

5. Notice that for $k = 0$ and $k = 1$, the terms become exactly zero. Therefore, we can shift the summation index to start from $k = 2$ and cancel out the terms in the denominator factorial: $k! = k(k-1)(k-2)!$

$$E[X(X-1)] = \sum_{k=2}^{\infty} k(k-1) \frac{e^{-\lambda} \lambda^k}{k(k-1)(k-2)!} = e^{-\lambda} \sum_{k=2}^{\infty} \frac{\lambda^k}{(k-2)!}$$

6. Factor out λ^2 from the numerator term to align the powers with the factorial index:

$$E[X(X-1)] = \lambda^2 e^{-\lambda} \sum_{k=2}^{\infty} \frac{\lambda^{k-2}}{(k-2)!}$$

7. Let $m = k - 2$. As k ranges from 2 to ∞ , m ranges from 0 to ∞ . The summation reveals a standard Taylor series expansion for the exponential function:

$$\sum_{m=0}^{\infty} \frac{\lambda^m}{m!} = e^{\lambda}$$

8. Substitute this back into our expression: $E[X(X-1)] = \lambda^2 e^{-\lambda} \cdot e^{\lambda} = \lambda^2$.

9. Reassemble the raw second moment: $E[X^2] = \lambda^2 + \lambda$.

10. Finally, compute the variance: $\text{Var}(X) = (\lambda^2 + \lambda) - (\lambda)^2 = \lambda$. The theorem is proved.

Section 4.2: Continuous Distribution Problems

Problem 2: Probability Mapping Over Continuous Uniform Scales

A continuous random variable X is uniformly distributed over the interval $[10, 30]$.

- (i) Write down its probability density function $f(x)$.
- (ii) Calculate the probability that X falls between 15 and 22, i.e., $P(15 \leq X \leq 22)$.

Detailed Step-by-Step Solution:

Part (i): Finding the Density Profile $f(x)$

1. For a uniform distribution across bounds $[a, b]$, the density function is a flat constant: $f(x) = \frac{1}{b-a}$.
2. Substitute the given boundary values $a = 10$ and $b = 30$:

$$f(x) = \frac{1}{30 - 10} = \frac{1}{20} \quad \text{for } 10 \leq x \leq 30$$

Part (ii): Computing the Target Interval Probability $P(15 \leq X \leq 22)$

1. Integrate the density function across the specified limits:

$$P(15 \leq X \leq 22) = \int_{15}^{22} \frac{1}{20} \, dx$$

2. Since the integrand is a constant, the integral simply scales with the width of the sub-interval:

$$P(15 \leq X \leq 22) = \frac{1}{20} \Big[x \Big]_{15}^{22} = \frac{1}{20} (22 - 15)$$

$$P(15 \leq X \leq 22) = \frac{7}{20} = 0.35$$

Conceptual Insight: Geometrically, the uniform distribution forms a rectangle of base length 20 and height $1/20$. The required probability is simply the area of the sub-rectangle stretching from 15 to 22, which has a base width of $22 - 15 = 7$ and an identical height of $1/20$, yielding exactly $7/20$.

Problem 3: Standard Normal Area Integration (Z-Score Matrix)

The operational lifespan of a critical component inside an automated industrial system follows a Normal distribution with a mean lifespan of $\mu = 500$ days and a standard deviation of $\sigma = 40$ days. Calculate the proportion of components that are expected to fail before reaching 450 days of operation. (Assume standard Z-table area value for $\Phi(-1.25) = 0.1056$).

Detailed Step-by-Step Solution:

1. Identify the core parameters of the continuous normal space: $\mu = 500$ and $\sigma = 40$.
2. We need to find the probability that the random variable X is less than 450: $P(X < 450)$.
3. Transform the raw scale coordinate into standard Normal score format (Z-score) to find how many standard deviations 450 lies away from the mean:

$$Z = \frac{X - \mu}{\sigma} = \frac{450 - 500}{40} = \frac{-50}{40} = -1.25$$

4. Rewrite the probability statement in terms of the standard normal variable Z :

$$P(X < 450) = P(Z < -1.25)$$

5. The cumulative area under the standard normal curve from negative infinity to a point z is given by the notation $\Phi(z)$. Using the provided standard normal distribution values:

$$P(Z < -1.25) = \Phi(-1.25) = 0.1056$$

6. Thus, approximately **10.56%** of the components are mathematically expected to fail before completing 450 operational days.